

Multi-Objective Alignment of LLMs: Literature Audit and a Proposed Research Direction (Review memo accompanying the RONPO draft, `main.tex`)

Advisor memo (K. Kim)

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TL;DR

RONPO’s exact slot (training-time maxmin over judge–response pairs with a last-iterate rate) is technically still vacant, but it is vacant because it is a Cartesian-product lifting of known machinery — the draft’s own [CRITICAL] comment — and everything around it got occupied in the last 12 months. The high-contrast direction recommended here: **alignment as a cooperative bargain among adversarial judges** — define, axiomatize, and provably reach the *Nash-bargaining / Kalai–Smorodinsky point* of the feasible set of per-judge worst-case win-rate floors, with the reference policy as the disagreement point. Four parallel web audits verified that *no* paper uses a cooperative bargaining solution as the training-time objective or Pareto-selection rule over multiple judges for a single LLM policy, *no* Kalai–Smorodinsky + RLHF work exists at all, and *no* axiomatic treatment of “which point on the multi-judge Pareto front should alignment pick” exists — but three adjacent papers appeared in the last six months, so this window is closing.

1 What the audit found (all web-verified as of 2026-07-20)

The convergence race is over. EGPO (Zhou, Fazel, Du; COLM 2025, arXiv:2503.08942) and Nash-MP (Tiapkin et al., arXiv:2505.19731) both give *linear* last-iterate convergence for KL-regularized NLHF, including noisy feedback. RONPO’s $O(1/T)$ OMD rate is not just dated, it is two generations behind. Competing on rates is a dead end.

The neighboring formulations are taken. MNPO (ICLR 2026, arXiv:2509.23102) has the per-player-oracle formulation but explicitly no heterogeneous theory. RMOD (ICLR 2026, arXiv:2503.08796) owns decoding-time maxmin over objective weights. Latham–Moharrami (arXiv:2604.00200) own offline constrained multi-oracle finite-sample theory. One correction to the inline comment in the draft: **MOPO (ICML 2026, arXiv:2505.10892) does not have adversarial lower bounds** — its “lower bounds” are user-set constraint thresholds in an ε -constraint program; there is no adversary over objectives. That actually weakens MOPO as a threat but also removes it as the strawman the comment implied.

The vector-payoff direction initially favored is half-taken. The “vector von Neumann winner” already exists under the name *Blackwell winner* (Bhatia, Pananjady, Bartlett, Dragan, Wainwright; NeurIPS 2020), and PROSPER (arXiv:2602.19041, Feb 2026, from the Swamy/Wu line) already fine-tunes LLMs toward a maximum-entropy Blackwell winner for cyclic multi-objective preferences. An approachability paper now would be a follow-up, not an opening move.

The social-choice layer is crowded at the top, thin at the center. Position papers, axioms for reward learning (Ge et al., NeurIPS 2024), Condorcet-cycle impossibility (Liu et al., *Annals of Statistics*, arXiv:2503.10990), distortion analysis showing NLHF is minimax-optimal (arXiv:2505.23749), population-proportional alignment (ICLR 2026, arXiv:2506.05619) — all taken. What nobody axiomatizes is the *selection problem*: given K conflicting judges, which single point on the feasible trade-off surface should the one deployed policy target, and why.

The bargaining check — the decisive one. Nash *welfare* aggregation over learned scalar reward functions exists as pure theory with no LLM experiments (Zhong et al., arXiv:2403.05006; Park et al., arXiv:2405.00254). MaxMin-RLHF (ICML 2024) is the egalitarian corner, never framed as bargaining. DISCA (arXiv:2605.10843, May 2026) name-drops Nash and Kalai–Smorodinsky for inference-time cultural persona blending but deploys a utilitarian mixture. Nash-MTL (ICML 2022) does gradient-space bargaining for multi-task vision. Full-text arXiv queries return **zero** papers combining “bargaining” with RLHF or preference optimization, and exactly **zero** hits for Kalai–Smorodinsky anywhere near RLHF. The citation graphs of MaxMin-RLHF, NLHF, and Panacea contain no bargaining follow-up. The center is open.

2 The proposal: Bargained Preference Optimization

Setup. K general (possibly intransitive) preference oracles P_k — exactly the NLHF setting the student already has. For a policy π , define its *floor* under judge k as

$$v_k(\pi) = \min_a P_k(\pi \succ a),$$

the worst-case win rate against an adversarial response. Each v_k is a minimum of linear functions, hence concave, so the feasible floor set

$$F = \{u : u \leq v(\pi) \text{ for some } \pi\}$$

is compact and convex. This is the object the whole multi-objective alignment literature is implicitly arguing about without naming it:

- **Utilitarian corner** (maximize $\sum_k v_k$) $\approx$ SPPO-avg / INPO-avg — the averaging baselines.
- **Egalitarian corner** (maximize $\min_k v_k$) = exactly RONPO at $\kappa \rightarrow 0$, and MaxMin-RLHF’s rule.
- A clean structural fact: since any w -weighted mixture of skew-symmetric oracles is skew-symmetric, *every scalarized preference game has value exactly $\frac{1}{2}$* . So the ideal point $(\frac{1}{2}, \dots, \frac{1}{2})$ is canonical, F touches it iff the judges share a common maximin winner, and the Condorcet-cycle results of Liu et al. imply that generically it doesn’t. The entire paper is one picture: the floor polytope F , its unreachable ideal corner, and where each existing method lands on it.

The contribution: don’t pick a corner — bargain. Treat the K judges as bargaining parties, with disagreement point $d = v(\mu)$, the floors of the *reference policy*. (In negotiation terms, the reference model is the BATNA — the outcome each judge gets if alignment “fails.”) Then:

- The **Nash bargaining solution** maximizes $\sum_k \log(v_k(\pi) - d_k)$. This is log-of-concave, hence a *concave program* — no saddle, no general-sum trap, unique optimum in floor space, unique policy with the τ -KL term.

- The **KS solution** maximizes $\min_k (v_k(\pi) - d_k) / (u_k^* - d_k)$ — a maximin over *canonically normalized surpluses*. This directly kills the min-max-normalization arbitrariness flagged in the draft (comment near the deployed-surrogate section): the normalization is no longer a pool-dependent hack, it is the axiomatically mandated scale, with ideal points u_k^* being (estimable) per-judge game values.

Why this is high-contrast rather than another aggregator. Three claims none of the occupied works make:

1. **An axiomatic uniqueness theorem on the right domain.** On the class of floor sets induced by K -judge preference games, NBS is the unique selection rule satisfying Pareto optimality, symmetry, scale covariance of surpluses, and IIA; KS is the unique one swapping IIA for monotonicity. This needs a *realizability lemma* — every compact convex comprehensive set in the relevant box is (approximately) the floor set of some K -judge game — which is genuine, non-lifting technical content, exactly what the draft’s central novelty comment demanded.
2. **An individual-rationality theorem with teeth.** At the NBS/KS optimum every judge’s worst-case win rate strictly exceeds its reference floor. Averaging provably violates this — and the violation *is* the hidden-objective collapse the student’s decoy game already demonstrates. The synthetic experiment gets upgraded from anecdote to axiom-violation certificate, with the existing decoy construction reused as the separating instance.
3. **Valid certificates, finally.** The draft’s [CRITICAL] comment on the softmin direction has a constructive fix: the *sum-form* log-sum-exp,

$$-\kappa \log \sum_i e^{-c_i/\kappa}, \quad \min_i c_i - \kappa \log n \leq -\kappa \log \sum_i e^{-c_i/\kappa} \leq \min_i c_i,$$

is a true *lower* bound on the minimum (sandwiched within $\kappa \log n$ below it), unlike the expectation form the draft used. Optimizing $\sum_k \log(\text{softmin} - d_k)$ with sum-form smoothing means every reported floor is a genuine certificate, and a Hoeffding-plus-union-bound argument gives the finite-pool, finite-query generalization bound the draft’s comments asked for — with probability $1 - \delta$, every judge’s true floor is at least the reported value minus $O(\sqrt{\log(K|\mathcal{A}|/\delta)/n})$.

Algorithm — and why the student can execute it. The NBS gradient is $\sum_k w_k \nabla v_k$ with $w_k \propto 1/(v_k - d_k)$: *inverse-surplus weighting*. Judges closest to their disagreement floor shout loudest; no adversary temperature κ as a hyperparameter — the weights are determined by measured surpluses. The inner minimization per judge is over the finite pool (closed-form Gibbs under smoothing), and the policy update remains exactly the partition-free log-ratio regression from INPO that the student already implemented. The deployed change relative to the current codebase is: replace the entropic (k, a) -adversary with per-judge worst-case opponents and inverse-surplus weights, and report per-judge surplus $v_k(\pi) - v_k(\mu)$ as the primary metric — which also resolves the theory-metric-mismatch comment, because the evaluation metric now *is* the theoretical object. Convergence: smoothed strongly-concave program $\Rightarrow$ last-iterate rates by standard means; not the paper’s selling point, so being “only” $O(1/T)$ or linear-under-smoothing is fine.

One more experiment nobody has run (verified untaken). The *panel-level* Condorcet structure of real judge panels — take Skywork/Athene/ArmoRM majority relations over actual

generation pools and measure cycles and Condorcet-winner existence. Existing work measures within-judge intransitivity only. The student’s pools make this a week of work, and it is the empirical motivation that makes reward-representability impossibility concrete: when the panel’s majority relation has cycles, no scalarization is faithful, and floors + bargaining is the natural response.

3 Threats and pre-emptive positioning

Closest work	Why it doesn’t cover this
Zhong et al. arXiv:2403.05006 (Nash welfare, multi-party RLHF)	Aggregates learned scalar <i>utilities</i> ; theory-only, no LLM experiments; no adversarial floors, no reference-policy disagreement point, no axiomatic characterization on preference-game domains
MaxMin-RLHF / RONPO-as-drafted	The egalitarian corner of our F — subsumed and axiomatically diagnosed (violates scale covariance of surpluses)
PROSPER (Blackwell winner)	Picks a <i>target set</i> by hand; no answer to which point, no axioms, no certificates
DISCA	Inference-time persona logit blending; bargaining is cited, not used
Nash-MTL	Gradient-space heuristic in vision MTL; no outcome-space solution concept, no oracles, no guarantees

Referee attack to prepare for: “bargaining among AI judges is anthropomorphic.” Answer: the axioms are design desiderata for an aggregation rule (scale invariance = robustness to judge recalibration; IR = no-judge-left-behind; IIA/monotonicity = stability under pool changes), not claims about negotiating humans. A wrinkle to handle honestly: NBS needs strictly positive surplus to be well-defined; if μ is already Pareto-optimal in floor space, fall back to leximin — one lemma, standard in bargaining theory.

4 Practical plan

AAAI-27’s deadline is roughly two weeks away — a pivot is not executable by then, and I’d advise against splitting the student’s attention with a salvage submission of RONPO, whose novelty problem the draft’s comments already establish is structural. The realistic target is **ICLR 2027 (abstract ~mid-September, ~2 months)**: infrastructure reuse is near-total (pools, judges, INPO-style trainer, decoy game, SafeRLHF setup), the new theory (realizability lemma, axiomatization, IR theorem, certificate bounds) is self-contained convex analysis plus classical bargaining theory — well matched to a student who needs the math to be checkable — and the panel-cycle study is a cheap, novel motivation section. AISTATS 2027 (~October) is the fallback if the LLM-scale runs lag: the theory + synthetic package stands alone there.

Runner-up direction, if a second arrow is wanted. Online/dueling-regret algorithms for Blackwell winners — multi-objective *dueling* bandits are confirmed empty (the MO-bandit literature is all numeric vector rewards), and PROSPER left the online/exploration side untouched. That’s a clean COLT/AISTATS theory paper, but it is a follow-up to Bhatia et al. and PROSPER rather than territory of your own, which is why it is ranked second.

The single sentence for the student: *we stop asking “how do we make the worst judge happy” and prove, axiomatically, what the K judges would jointly agree the one deployed model should be — then train to that point with certificates that are actually valid.*